

Unit 2: Number Operations

Finding, recalling and using addition and subtraction facts up to 20

Addition means combining things to get total

Example: Count and add.



4 apples

and



3 apples

makes



7 apples

We write it as

$$4 + 3 = 7$$

and read as

Four plus three is equal to seven

Where,

"+" is the sign of addition

and "=" is the sign of equality.



Activity

Count and add.



3 balloons

+



5 balloons

=



8 balloons

3

+

5

=

8



__ bananas + __ bananas = __ bananas

$$5 + 6 = 11$$



__ Stars + __ Stars = __ Stars

$$\square + \square = \square$$

These are
some addition
facts

$$\begin{aligned} 4 + 3 &= 7 \\ 3 + 5 &= 8 \\ 5 + 6 &= 11 \\ 6 + 8 &= 14 \end{aligned}$$

Unit 2: Number Operations

Some other addition facts up to 20 are

Addition of 1

$$\begin{array}{l} 1 + 1 = 2 \\ 2 + 1 = 3 \\ 3 + 1 = 4 \\ 4 + 1 = 5 \\ 5 + 1 = 6 \end{array}$$

Addition of 2

$$\begin{array}{l} 1 + 2 = 3 \\ 2 + 2 = 4 \\ 3 + 2 = 5 \\ 4 + 2 = 6 \\ 5 + 2 = 7 \end{array}$$

Addition of 3

$$\begin{array}{l} 1 + 3 = 4 \\ 2 + 3 = 5 \\ 3 + 3 = 6 \\ 4 + 3 = 7 \\ 5 + 3 = 8 \end{array}$$

Addition of 4

$$\begin{array}{l} 6 + 4 = 10 \\ 7 + 4 = 11 \\ 8 + 4 = 12 \\ 9 + 4 = 13 \end{array}$$

Addition of 5

$$\begin{array}{l} 6 + 5 = 11 \\ 7 + 5 = 12 \\ 8 + 5 = 13 \\ 9 + 5 = 14 \end{array}$$

Addition of 6

$$\begin{array}{l} 6 + 6 = 12 \\ 7 + 6 = 13 \\ 8 + 6 = 14 \\ 9 + 6 = 15 \end{array}$$

Addition of 7

$$\begin{array}{l} 7 + 7 = 14 \\ 8 + 7 = 15 \\ 9 + 7 = 16 \\ 10 + 7 = 17 \end{array}$$

Addition of 8

$$\begin{array}{l} 7 + 8 = 15 \\ 8 + 8 = 16 \\ 9 + 8 = 17 \\ 10 + 8 = 18 \end{array}$$

Addition of 9

$$\begin{array}{l} 7 + 9 = 16 \\ 8 + 9 = 17 \\ 9 + 9 = 18 \\ 10 + 9 = 19 \end{array}$$

Addition of 10

$$\begin{array}{l} 7 + 10 = 17 \\ 8 + 10 = 18 \\ 9 + 10 = 19 \\ 10 + 10 = 20 \end{array}$$

Subtraction means taking away

Example:

2 eggs are broken from 6 eggs how many are left?



Total eggs	6
Broken eggs	2
Remaining eggs	4

We write it as

$$6 - 2 = 4$$

Where

"-" is the symbol of subtraction.



Activity

Complete the following



Total parrots	Flew away	Remaining
5	2	



Activity

Complete the following



Total lemon	Taken away	Remaining

These are
some
subtraction
facts

$$6 - 2 = 4$$

$$5 - 2 = 3$$

$$8 - 3 = 5$$

Some other addition and subtraction facts:

$$8 + 2 = 10$$

$$9 + 6 = 15$$

$$8 + 9 = 17$$

$$12 + 8 = 20$$

$$10 - 8 = 2$$

$$15 - 6 = 9$$

$$17 - 9 = 8$$

$$20 - 8 = 12$$



Activity

Complete the following.

$$7 + 4 = 11$$

$$6 + 3 = \square$$

$$14 + 6 = \square$$

$$17 + 2 = \square$$

$$11 - 7 = 4$$

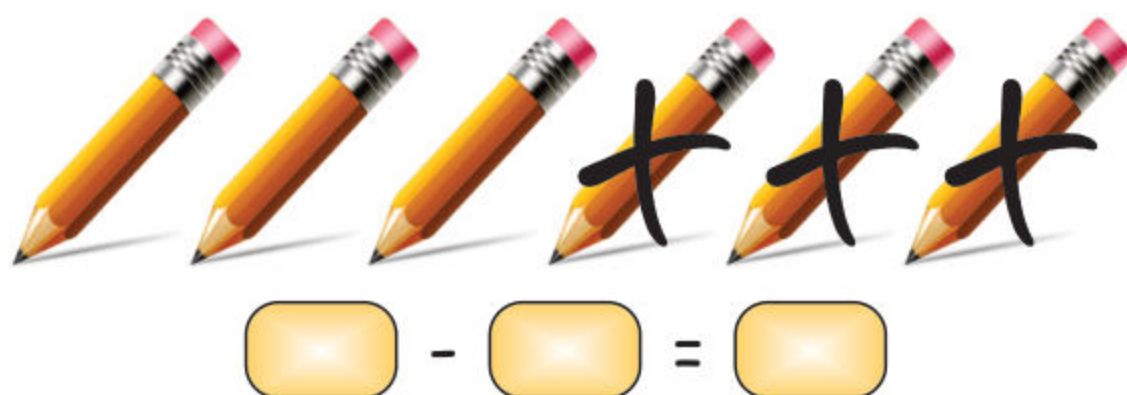
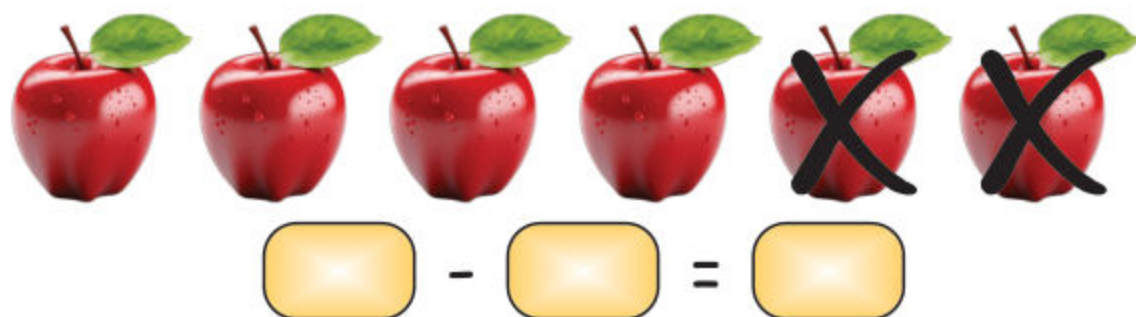
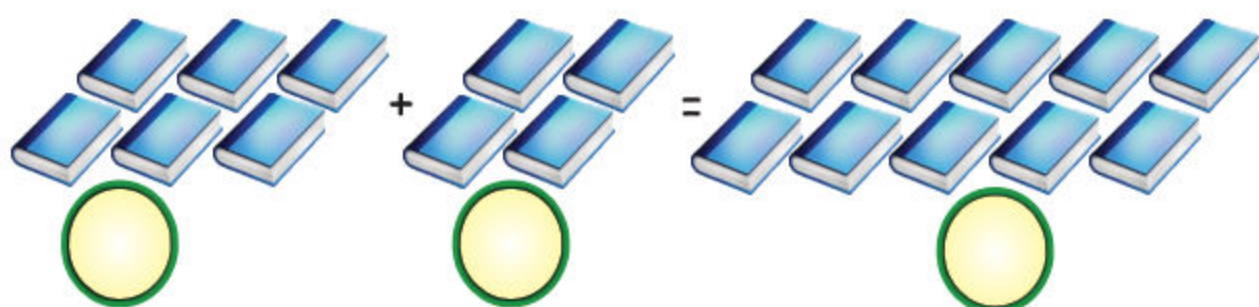
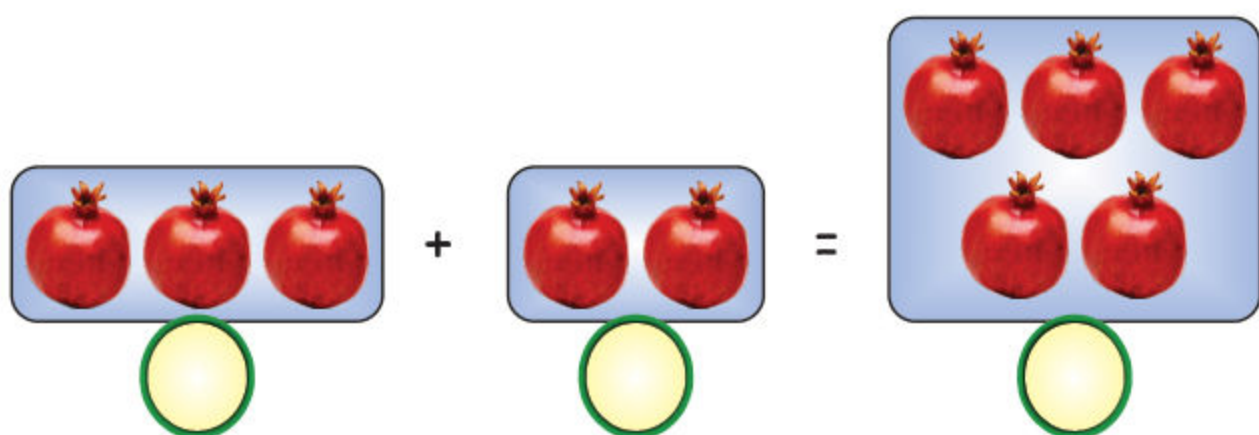
$$9 - 6 = \square$$

$$20 - 14 = \square$$

$$19 - 2 = \square$$

Exercise 1

1. Complete the following



2. Match the following.

$5 + 7$	14
$8 + 6$	12
$9 + 8$	6
$8 - 3$	17
$10 - 4$	16
$18 - 2$	5

3. Complete the following addition and subtraction facts.

$2 + 5 = 7$	$17 - 5 = 12$
$9 + 8 = \square$	$15 - 8 = \square$
$16 + 4 = \square$	$20 - 16 = \square$
$7 + \square = 13$	$13 - \square = 6$
$11 + \square = 20$	$20 - \square = 11$

4. Match the following.

$5 + 7$	$3 + 17$
$13 + 4$	$4 + 15$
$17 + 3$	$6 + 8$
$8 + 6$	$7 + 5$
$15 + 4$	$4 + 13$

Comparing numbers

Lets us compare numbers to find how many more or less by the following activities.



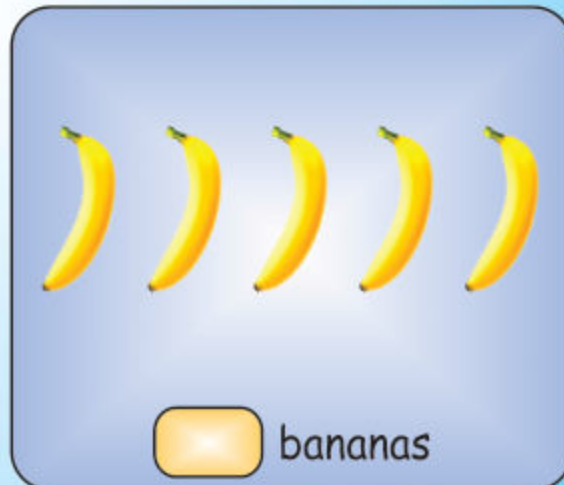
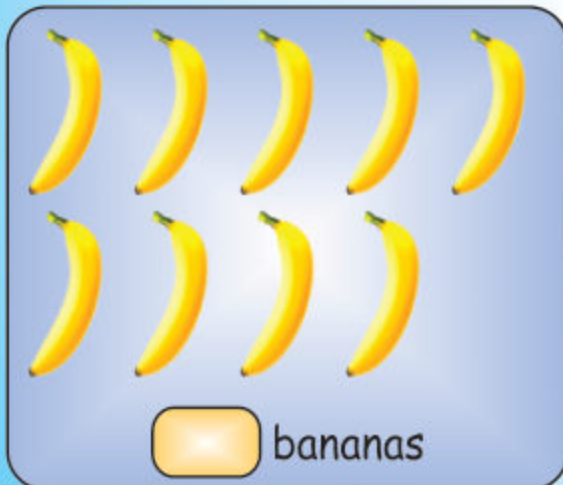
Activity

Count and write the number. Also tick (✓) the number of more objects and find how many more.



$$8 - 5 = 3$$

8 is more than 5 by 3



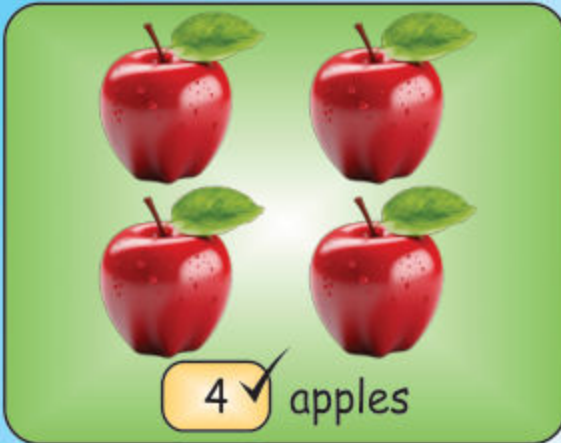
$$9 - 5 = 4$$

___ is more than ___ by ___



Activity

Count and write the number. Also tick (✓) the number of less objects and find how many less.



$$6 - 4 = 2$$

4 is less than 6 by 2



$$7 - 3 = 4$$

___ is less than ___ by ___

Exercise 2

- Count and write the number. Tick the number of more objects and find how many more.



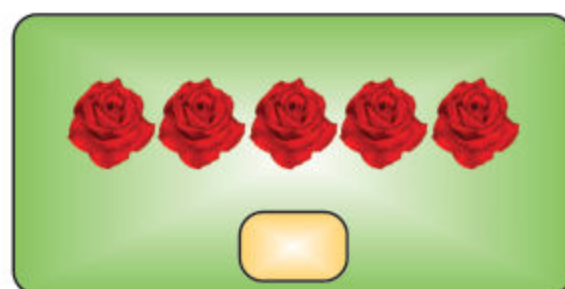
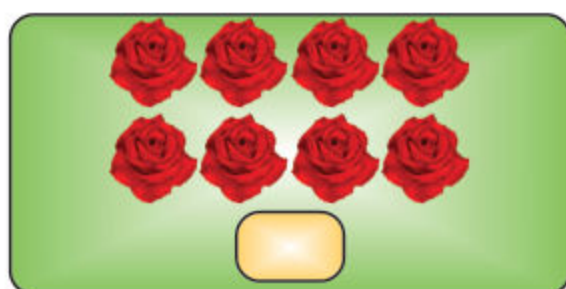
$$6 - 4 = 2$$

6 is more than 4 by 2



$$\text{ } - \text{ } = \text{ }$$

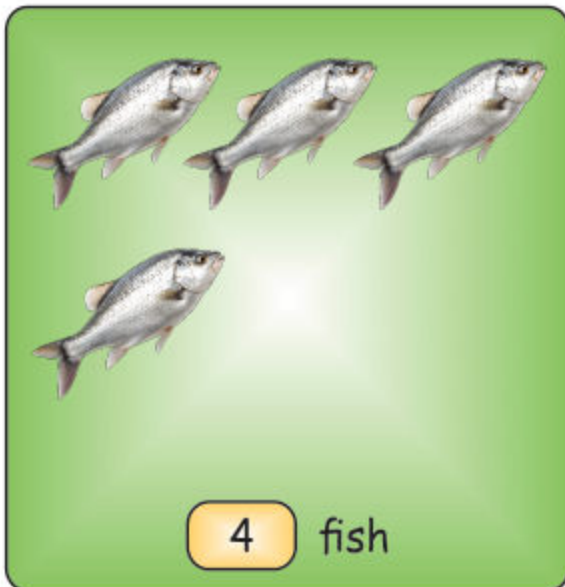
__ is more than __ by __



$$\text{ } - \text{ } = \text{ }$$

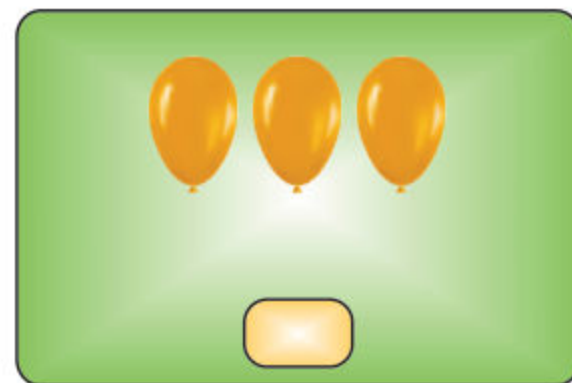
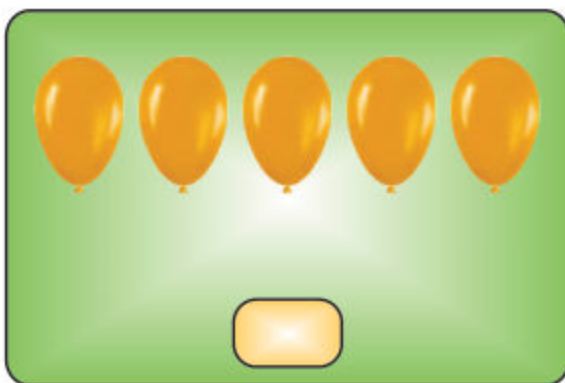
__ is more than __ by __

2. Count and write the number. Tick the number of less objects and find how many less.



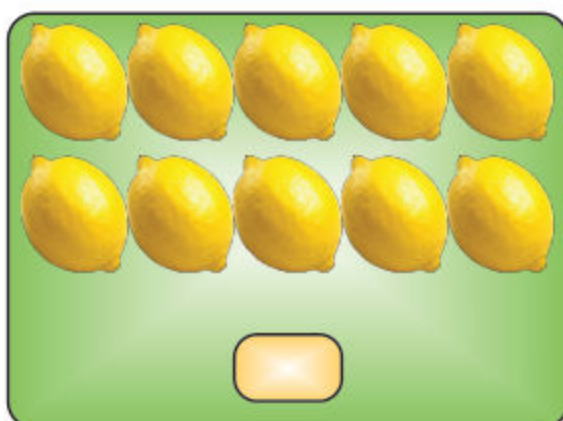
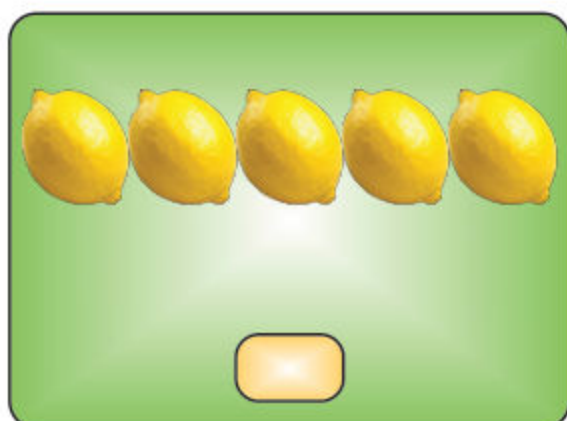
$$7 - 4 = 3$$

4 is less than 7 by 3



$$\text{ } - \text{ } = \text{ }$$

__ is less than __ by __



$$\boxed{} - \boxed{} = \boxed{}$$

$\boxed{}$ is less than $\boxed{}$ by $\boxed{}$

Addition of 1-digit numbers

We can add 1-digit numbers by using addition facts or by just counting together.

Example Add the following numbers mentally.

$$\boxed{5 + 2 = 7}$$

$$\boxed{7 + 3 = 10}$$

$$\boxed{3 + 0 = 3}$$

$$\boxed{4 + 9 = 13}$$

Example Add mentally.

$$\boxed{2 + 3 + 6 = 11}$$

Explanation:

First we add any two numbers then add the result with the remaining number.

$$\begin{array}{c} 2 + 3 + 6 = 11 \\ \quad \swarrow \quad \searrow \\ \quad \quad 5 \end{array}$$



Activity

Add the following mentally.

$$6 + 2 = \boxed{8}$$

$$5 + 0 = \boxed{}$$

$$5 + 7 = \boxed{}$$



Activity

Add the following mentally.

$$6 + 4 + 5 = \boxed{15}$$

10

$$7 + 4 + 2 = \boxed{}$$

11

$$5 + 6 + 3 = \boxed{}$$

11

Longhand addition: In long hand addition, we add numbers vertically by aligning the place values as explained in the following example

Example Add 6 and 8 by long hand addition

$$\begin{array}{r} 6 \\ + 8 \\ \hline 14 \end{array}$$

Explanation:

Ten	One
-----	-----

$$\begin{array}{r} 6 \\ + 8 \\ \hline \end{array}$$

Write numbers in a consistent order.



Ten	One
-----	-----

$$\begin{array}{r} 6 \\ + 8 \\ \hline 4 \end{array}$$

1

$6+8=14$
Write only 4 in the ones place. For 10, write a small 1 in the tens place.



Ten	One
-----	-----

$$\begin{array}{r} 6 \\ + 8 \\ \hline 1 \quad 4 \end{array}$$

1

Calculate the tens place. Since it is just 1 carried over, write 1 in the tens place.



Activity

Add the following.

$$\begin{array}{r} 7 \\ + 5 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 9 \\ + 8 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 4 \\ + 5 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 3 \\ + 9 \\ \hline \\ \hline \end{array}$$

$$\begin{array}{r} 4 \quad \curvearrowright 9 \\ 5 \quad \curvearrowleft \\ + 3 \\ \hline 12 \\ \hline \end{array}$$

$$\begin{array}{r} 6 \quad \curvearrowright 10 \\ 4 \quad \curvearrowleft \\ + 8 \\ \hline \\ \hline \end{array}$$

Mental addition of 2-digit numbers with tens

Example

Add the following mentally:

$$\boxed{54} + \boxed{20} = \boxed{74}$$

We just add ones mentally i.e. $4 + 0 = 4$

and then add tens mentally i.e. $5 + 2 = 7$

Similarly

$$\boxed{75} + \boxed{23} = \boxed{98}$$

Addition of 2-digit numbers with tens in written form

We add respective place values for addition as shown below

Example

Add the following:

	T	O
	3	4
+	5	0
	8	4

	T	O
	2	7
+	6	0
	8	7

	T	O
	5	6
+	2	0
	7	6

Subtraction of 1-digit numbers

We can easily subtract 1-digit numbers by using subtraction facts as done in the following examples.

Example

Perform the subtraction mentally.

$$\boxed{6 - 3 = 3}$$

$$\boxed{12 - 4 = 8}$$

$$\boxed{9 - 5 = 4}$$

$$\boxed{8 - 7 = 1}$$

Example

Perform the subtraction in written form.

$$\begin{array}{r} 8 \\ - 5 \\ \hline 3 \end{array}$$

$$\begin{array}{r} 7 \\ - 2 \\ \hline 5 \end{array}$$

$$\begin{array}{r} 6 \\ - 3 \\ \hline 3 \end{array}$$

Subtraction of tens from 2-digit numbers mentally

In order to subtract tens from 2-digit numbers mentally, we only subtract ones from ones and tens from tens as done in the following example.

Example

Perform the subtraction mentally.

$$46 - 30 = 16$$

$$57 - 20 = 37$$

$$64 - 40 = 24$$

$$75 - 20 = 55$$

Subtraction of tens from 2-digit numbers in written form

We subtract respective place values for subtraction of tens from 2-digit numbers in written form as shown in the following example.

Example

Subtract the following.

$$\begin{array}{r} \text{T O} \\ 84 \\ - 60 \\ \hline 24 \end{array}$$

$$\begin{array}{r} \text{T O} \\ 95 \\ - 30 \\ \hline 65 \end{array}$$

$$\begin{array}{r} \text{T O} \\ 76 \\ - 10 \\ \hline 66 \end{array}$$

Exercise 3

1. Add and subtract mentally.

$$\begin{array}{rcl} 5 & + & 4 = 9 \\ 8 & + & 7 = \square \\ 6 & + & 7 = \square \\ 9 & + & 0 = \square \\ 3 & + & 5 = \square \end{array}$$

$$\begin{array}{rcl} 9 & - & 3 = 6 \\ 8 & - & 5 = \square \\ 5 & - & 2 = \square \\ 7 & - & 6 = \square \\ 4 & - & 0 = \square \end{array}$$

2. Add and subtract mentally.

$$\begin{array}{rcl} 35 & + & 20 = 55 \\ 17 & + & 40 = \square \\ 64 & + & 30 = \square \\ 54 & + & 10 = \square \\ 27 & + & 50 = \square \end{array}$$

$$\begin{array}{rcl} 65 & - & 20 = 45 \\ 92 & - & 50 = \square \\ 78 & - & 40 = \square \\ 59 & - & 30 = \square \\ 48 & - & 30 = \square \end{array}$$

3. Add and subtract the following.

$\begin{array}{r} 5 \\ + 4 \\ \hline \end{array}$	$\begin{array}{r} 6 \\ + 2 \\ \hline \end{array}$	$\begin{array}{r} 36 \\ + 20 \\ \hline \end{array}$	$\begin{array}{r} 9 \\ - 4 \\ \hline \end{array}$	$\begin{array}{r} 8 \\ - 3 \\ \hline \end{array}$	$\begin{array}{r} 23 \\ - 10 \\ \hline \end{array}$
$\begin{array}{r} 47 \\ + 30 \\ \hline \end{array}$	$\begin{array}{r} 59 \\ + 40 \\ \hline \end{array}$	$\begin{array}{r} 65 \\ + 30 \\ \hline \end{array}$	$\begin{array}{r} 85 \\ - 30 \\ \hline \end{array}$	$\begin{array}{r} 77 \\ - 40 \\ \hline \end{array}$	$\begin{array}{r} 45 \\ - 20 \\ \hline \end{array}$
$\begin{array}{r} 28 \\ + 30 \\ \hline \end{array}$	$\begin{array}{r} 81 \\ + 50 \\ \hline \end{array}$	$\begin{array}{r} 73 \\ + 40 \\ \hline \end{array}$	$\begin{array}{r} 82 \\ - 20 \\ \hline \end{array}$	$\begin{array}{r} 45 \\ - 30 \\ \hline \end{array}$	$\begin{array}{r} 29 \\ - 10 \\ \hline \end{array}$

Mental Addition of a 2-digit number with 1-digit number

To add a 2-digit number with 1-digit number mentally, we only add ones of both numbers as explained in the following examples.

$$\begin{array}{c} \text{56} + \text{3} = \text{59} \quad | \quad \text{23} + \text{5} = \text{28} \end{array}$$

Addition of a 2-digit number with 1-digit number in written form

To add a 2-digit number with 1-digit number, we add place value of ones in written form as explained by the following example.

Example

T	O
5	2
+	4
5	6

T	O
9	4
+	3
9	7

T	O
6	7
+	2
6	9

Mental addition of two 2-digit numbers

For mental addition, we add ones with ones and tens with tens as explained in the following examples.

$$\begin{array}{c} \text{23} + \text{54} = \text{77} \quad | \quad \text{65} + \text{24} = \text{89} \end{array}$$

Addition of two 2-digit numbers in written form

To add two 2-digit numbers in written form, we only add respective place values as explained in the following examples.

T	O
5	2
+	3 6
8	8

T	O
6	4
+	2 5
8	9

T	O
8	6
+	1 2
9	8

Mental subtraction of a 2-digit number with 1-digit number

For this subtraction, we only subtract ones as explained in the following example.

$$\begin{array}{c} \text{56} \\ \text{3} \end{array} = \text{53} \quad \bigg| \quad \begin{array}{c} \text{75} \\ \text{4} \end{array} = \text{71}$$

Subtraction of a 2-digit number with 1-digit number in written form

For subtraction in written form we just subtract place value of ones as explained below

T	O
5	6
-	2
5	4

T	O
3	9
-	5
3	4

T	O
8	4
-	3
8	1

Mental subtraction of two 2-digit numbers

For mental subtraction we subtract ones from ones and tens from tens as explained below.

$$\begin{array}{c} \text{67} \\ \text{23} \end{array} = \text{44} \quad \bigg| \quad \begin{array}{c} \text{89} \\ \text{53} \end{array} = \text{36}$$

Subtraction of two 2-digit numbers in written form

For subtraction of two 2-digit numbers in written form, we subtract the respective place values as explained in the following examples.

T	O
8	7
-	2 3
6	4

T	O
9	5
-	6 3
3	2

T	O
4	7
-	2 3
2	4

Exercise 4

1. Add and subtract mentally.

$47 + 2 = \square$

$15 + 3 = \square$

$92 + 6 = \square$

$32 + 54 = \square$

$51 + 43 = \square$

$38 - 5 = \square$

$47 - 7 = \square$

$88 - 5 = \square$

$75 - 42 = \square$

$96 - 43 = \square$

3. Add and subtract the following.

TO

3 5

+ 3

TO

6 2

+ 7

TO

4 8

- 4

TO

9 9

- 8

TO

9 3

+ 4

TO

2 4

+ 3

TO

5 6

- 3

TO

7 9

- 6

TO

2 4

+ 6 3

TO

6 6

+ 2 3

TO

2 8

- 1 5

TO

4 3

- 2 1

TO

1 2

+ 8 5

TO

4 8

+ 3 0

TO

5 2

- 4 1

TO

6 8

- 3 5

Unit 2: Number Operations

Solve real-life word problems of addition and subtraction

Let us see some real - life problems of addition and subtraction.

Example 1

Hina has $\boxed{20}$ rupees
Her friend has $\boxed{15}$ rupees
 $+$
Total amount is $\boxed{35}$ rupees



Example 2

Moeen bought $\boxed{15}$ kg of flour
His son bought $\boxed{5}$ kg of flour
 $+$
Total quantity $\boxed{20}$ kg of flour



Example 3

Total number of students in a class = $\begin{array}{r} 29 \\ - 4 \\ \hline 25 \end{array}$
Number of absent students =
Number of present students =



Example 4

A teacher has two number cards.

Sum of numbers = $\begin{array}{r} 54 \\ - 34 \\ \hline 20 \end{array}$
If one card has number =
The missing number =



Exercise 5

1. Number of girls in a class = 16
Number of boys in the class = 23
Total number of students =

2. Abdul Rafay has = 36 rupees
Muhammad Taha has = 52 rupees
Their total amount =

3. Faiza bought 60 grams of almonds
Elsa bought 30 grams of almonds
Total weight = grams

4. A shopkeeper has = 46 eggs
He sold = 23 eggs
The number of remaining eggs =

5. The total cost of two toys = 95 rupees
Cost of one toy = 40 rupees
The cost of other toy =

6. Total number of parrots on a tree = 26
Number of parrots that flew = 12
The number of remaining parrots =

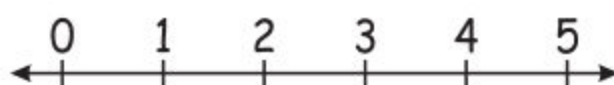
Estimation of the answer of addition and subtraction questions.

For estimation we will learn rounding off a number

Rounding a number means making the number simple

For rounding a number, we should know the number line.

A number line is the line with numbers as shown below.

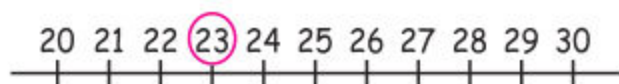


Note: Divisions must be equal on the line.

At this stage, we round numbers to the nearest ten as explained in the following examples.

Example Round off 23 to the nearest ten.

Solution:



20 is nearer to 23

So 23 is rounded to 20

we say 23 is rounded down to 20

i.e. 23 ↓ 20

Explanation:

1. First we, identify two tens. One ten before 23 and other after 23. Here 20 is before 23 and 30 is after 23.
2. Draw a number line indicating numbers from 20 to 30
3. See which ten is nearer to 23. Here 20 is nearer to 23. So 23 is rounded to 20

Rounding down



We ignore ones if they are less than 5 and keep the tens same.

Unit 2: Number Operations

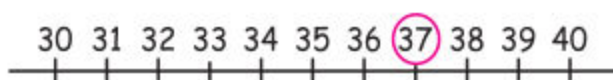
Some examples of rounding down are given below

Rounding down ↓

42	↓	40	64	↓	60
51	↓	50	83	↓	80

Example Round off 37 to the nearest ten

Solution:



40 is nearer to 37

So 37 is rounded to 40

We say 37 is rounded up to 40

i.e. 37 ↑ 40

Explanation:

1. First we, identify two tens. One ten before 37 and other after 37. Here 30 is before 37 and 40 is after 37.
2. Draw a number line indicating numbers from 30 to 40
3. See which ten is nearer to 37. Here 40 is nearer to 37. So 37 is rounded to 40

Rounding up



We ignore ones if they are 5 or more and increase one ten.

Some examples of rounding up are given below

Rounding up ↑

76	↑	80	39	↑	40
45	↑	50	8	↑	10

Estimation means getting approximate answer by rounding

Let us estimate answers of addition and subtraction by the following examples.

Example Estimate the sum of 46 and 21

Solution:

Here 46 ↑ 50
 and 21 ↓ 20

Now $50 + 20 = 70$

So Estimated sum = 70

$$\begin{array}{r} 50 \\ + 20 \\ \hline 70 \end{array}$$

Example Estimate the difference: $82 - 36$

Solution:

Here 82 ↓ 80
 and 26 ↑ 30

Now $80 - 30 = 50$

So The estimated difference = 50

$$\begin{array}{r} 80 \\ - 30 \\ \hline 50 \end{array}$$

Exercise 6

1. Match by rounding the numbers.

72	→	30
86		60
17		50
28		20
45		70
62		90

2. Estimate the following.

$$\begin{array}{rcl} 18 & + & 58 = \square \\ 32 & + & 49 = \square \\ 23 & + & 56 = \square \\ 27 & + & 61 = \square \\ 64 & + & 15 = \square \end{array}$$

$$\begin{array}{rcl} 46 & - & 21 = \square \\ 66 & - & 53 = \square \\ 84 & - & 17 = \square \\ 55 & - & 39 = \square \\ 78 & - & 22 = \square \end{array}$$

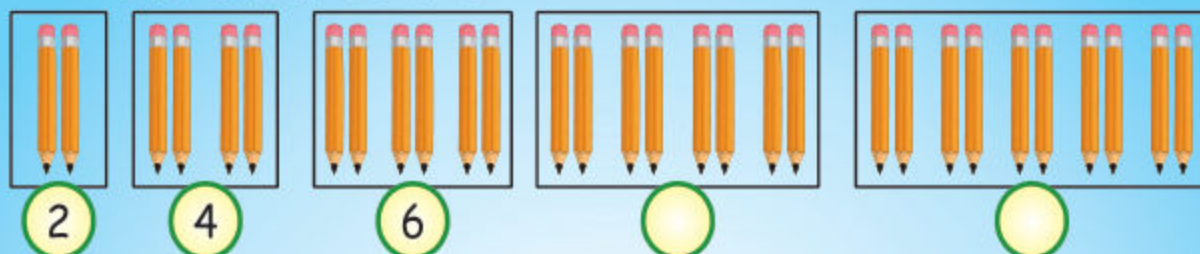
Counting and writing in 2s, 5s and 10s using concrete objects

Counting in 2s

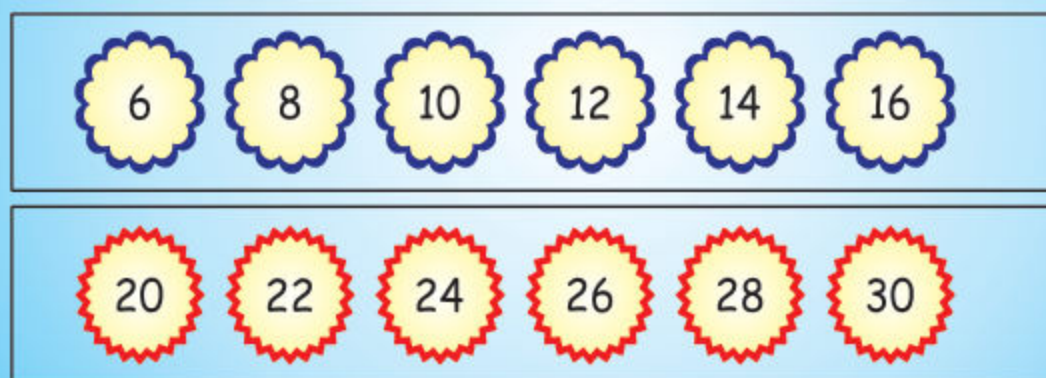


Activity

Count and write in 2s



Count in 2s



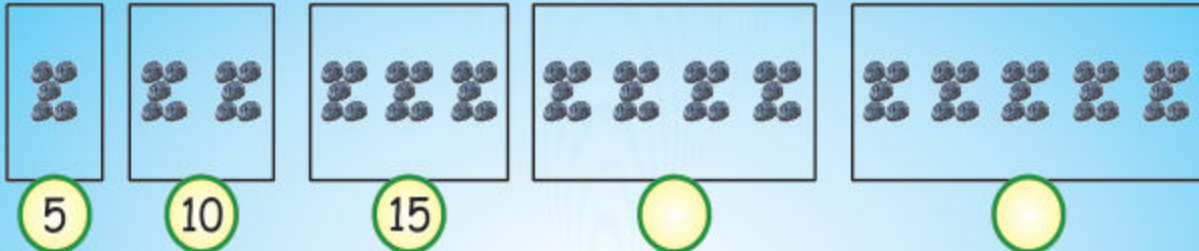
Count in 2s and write the missing numbers





Activity

Count and write in 5s.



Count in 5s



Count in 5s and write the missing numbers



Activity

Count and write in 10s



Count in 10s



Count in 5s and write the missing numbers



Exercise 7

1. Count in 2s and write the missing numbers.



2. Count in 5s and write the missing numbers.



3. Count in 10s and write the missing numbers.



4. Write the missing numbers in empty squares by count in 2s.

1		3		5		7		9	
11		13		15		17		19	
21		23		25		27		29	
31		33		35		37		39	
41		43		45		47		49	
51		53		55		57		59	
61		63		65		67		69	
71		73		75		77		79	
81		83		85		87		89	
91		93		95		97		99	

5. Count in 5s and write the missing numbers in empty squares.

1	2	3	4		6	7	8	9	
11	12	13	14		16	17	18	19	
21	22	23	24		26	27	28	29	
31	32	33	34		36	37	38	39	
41	42	43	44		46	47	48	49	
51	52	53	54		56	57	58	59	
61	62	63	64		66	67	68	69	
71	72	73	74		76	77	78	79	
81	82	83	84		86	87	88	89	
91	92	93	94		96	97	98	99	

6. Count in 2s and write the missing numbers.



7. Count in 5s and write the missing numbers.



8. Count in 10s and write the missing numbers.

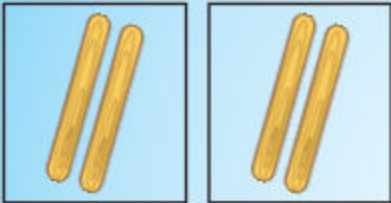


Recognizing counting in 2s, 5s and 10s as multiplication tables of 2, 5 and 10.



Activity

Count in 2s (twos) and write

Objects	Counting
	1 twos are 2
	2 twos are 4
	3 twos are 6

Similarly,

4 twos are 8
 5 twos are 10
 6 twos are 12
 7 twos are 14
 8 twos are 16
 9 twos are 18
 10 twos are 20

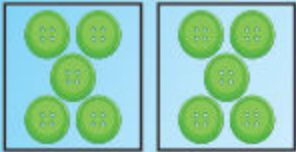
Multiplication table of 2

1	twos are	2
2	twos are	4
3	twos are	6
4	twos are	8
5	twos are	10
6	twos are	12
7	twos are	14
8	twos are	16
9	twos are	18
10	twos are	20



Activity

Count in 5s (fives) and write

Objects	Counting
	1 fives are 5
	2 fives are 10
	3 fives are 15

Similarly,

4	fives are	20
5	fives are	25
6	fives are	30
7	fives are	35
8	fives are	40
9	fives are	45
10	fives are	50

Multiplication table of 5

1	fives are	5
2	fives are	10
3	fives are	15
4	fives are	20
5	fives are	25
6	fives are	30
7	fives are	35
8	fives are	40
9	fives are	45
10	fives are	50



Activity

Count in 10s (tens) and write

Objects	Counting
	1 tens are 10
	2 tens are 20
	3 tens are 30

Similarly,

4	tens are	40
5	tens are	50
6	tens are	60
7	tens are	70
8	tens are	80
9	tens are	90
10	tens are	100

Note:

10 tens are called one hundred and we write it as 100

Multiplication table of 10

1	tens are	10
2	tens are	20
3	tens are	30
4	tens are	40
5	tens are	50
6	tens are	60
7	tens are	70
8	tens are	80
9	tens are	90
10	tens are	100

Exercise 8

1. Match the following.

4 twos	14
5 twos	16
6 twos	8
7 twos	10
8 twos	12

2. Match the following.

3 fives	20
4 fives	30
5 fives	15
6 fives	35
7 fives	25

3. Match the following.

2 tens	30
3 tens	20
4 tens	50
5 tens	60
6 tens	40

Recognizing multiplication as repeated addition

Let us count the number of wheels of the following tricycles.



We see each tricycle has 3 wheels.

So, we only add 3 repeatedly four times.

i.e $3 + 3 + 3 + 3 = 12$

or Four times 3 is 12

We write it as $4 \times 3 = 12$
and read it as "4 multiply 3 is equal to 12"

Note:

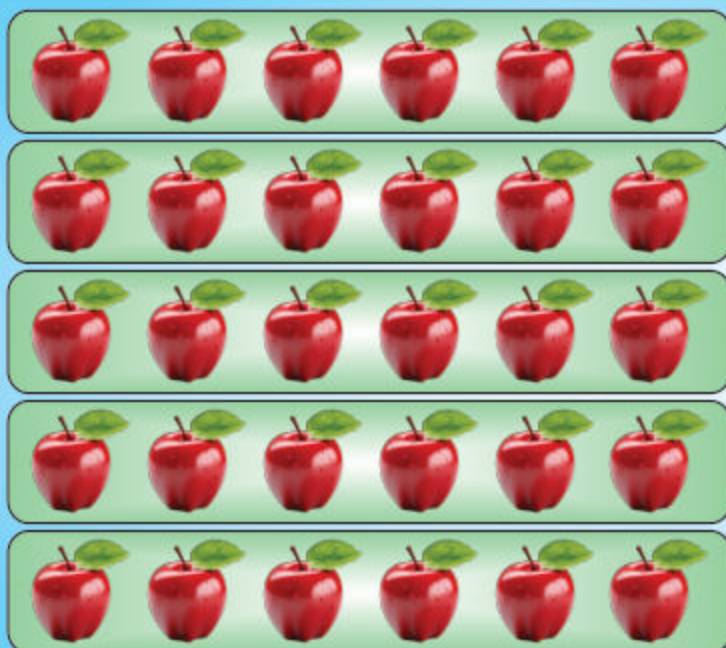
" $\times$ " is the sign of multiplication



Activity

Objects	Counting
	$2 + 2 + 2 = 6$ or $3 \text{ times } 2 = 6$ or $3 \times 2 = 6$
	$4 + 4 + 4 = 12$ or $3 \text{ times } 4 = 12$ or $3 \times 4 = 12$
	$5 + 5 + 5 + 5 = 20$ or $4 \text{ times } 5 = 20$ or $4 \times 5 = 20$

Unit 2: Number Operations



$$6 + 6 + 6 + 6 + 6 = 30$$

or $5 \text{ times } 6 = 30$

or $5 \times 6 = 30$



Activity

Match the following

$$7 + 7 + 7$$

$$8 + 8 + 8 + 8$$

$$2 + 2 + 2 + 2 + 2 + 2$$

$$4 + 4 + 4 + 4 + 4$$

$$5 + 5 + 5$$

$$6 \times 2$$

$$3 \times 5$$

$$4 \times 8$$

$$3 \times 7$$

$$5 \times 4$$

Exercise 9

1. Match the following.

$9 + 9 + 9$

$6 + 6 + 6 + 6$

$4 + 4 + 4 + 4 + 4$

$2 + 2 + 2 + 2 + 2$

$3 + 3 + 3 + 3 + 3 + 3$

5 times 4

6 times 3

3 times 9

4 times 6

5 times 2

2. Match the following.

7 times 2

9 times 5

4 times 6

8 times 3

5 times 4

9×5

8×3

7×2

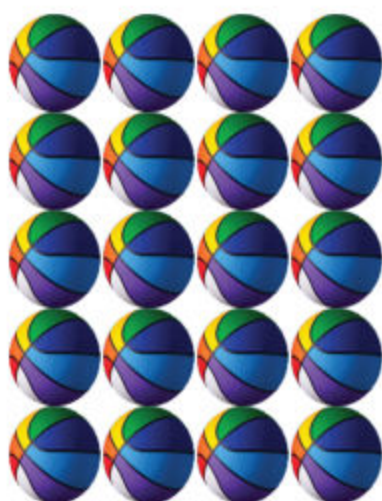
5×4

4×6

Recognizing that the multiplication of any two numbers can be done in any order.

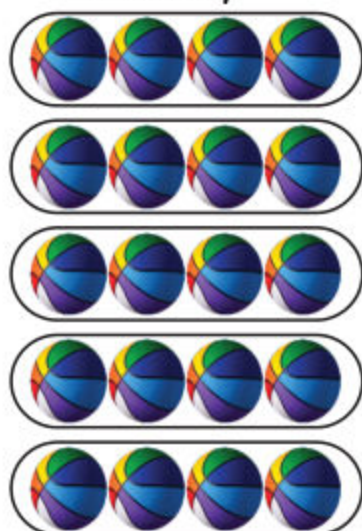
Example

Count the objects and write



We can count in two ways

First way



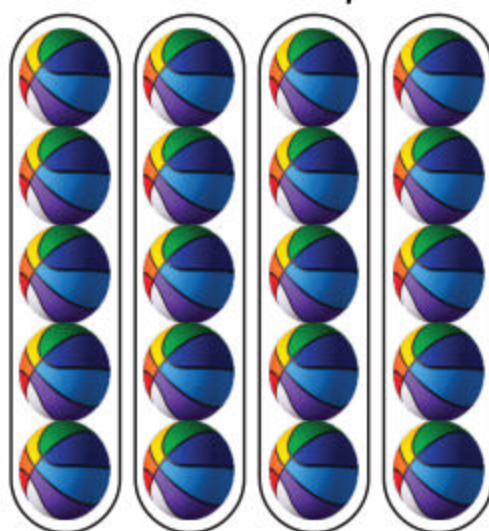
5 times 4

or 5×4

i.e. $5 \times 4 = 20 = 4 \times 5$

Similarly, $2 \times 6 = 6 \times 2$
 $3 \times 4 = 4 \times 3$

Second way



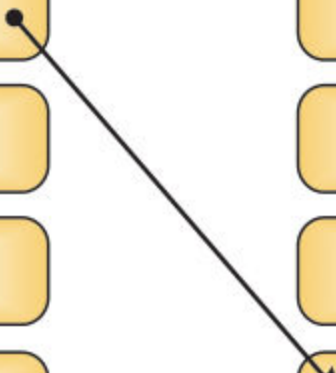
4 times 5

or 4×5

Exercise 10

1. Match the following.

5×6	4×6
3×7	5×8
6×4	2×9
8×5	6×5
9×2	7×3



2. Complete the following.

3	$\times$	8	=		$\times$	3
9	$\times$	4	=		$\times$	9
7	$\times$		=	6	$\times$	7
	$\times$	6	=	6	$\times$	2
5	$\times$	8	=	8	$\times$	

Recognizing division as repeated subtraction



Activity

Dividing 8 pieces of cake equally in the plates keeping 2 pieces of cake in each plate.



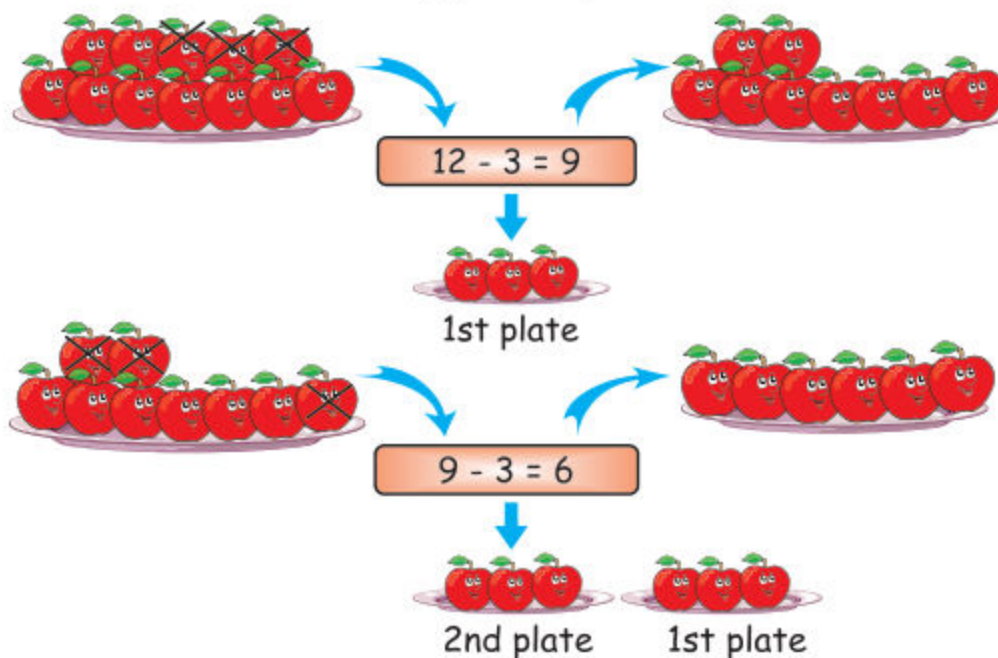
We see that 4 plates are needed. This is equal sharing of pieces of cake in 4 plates.

Division is the process of equal sharing.

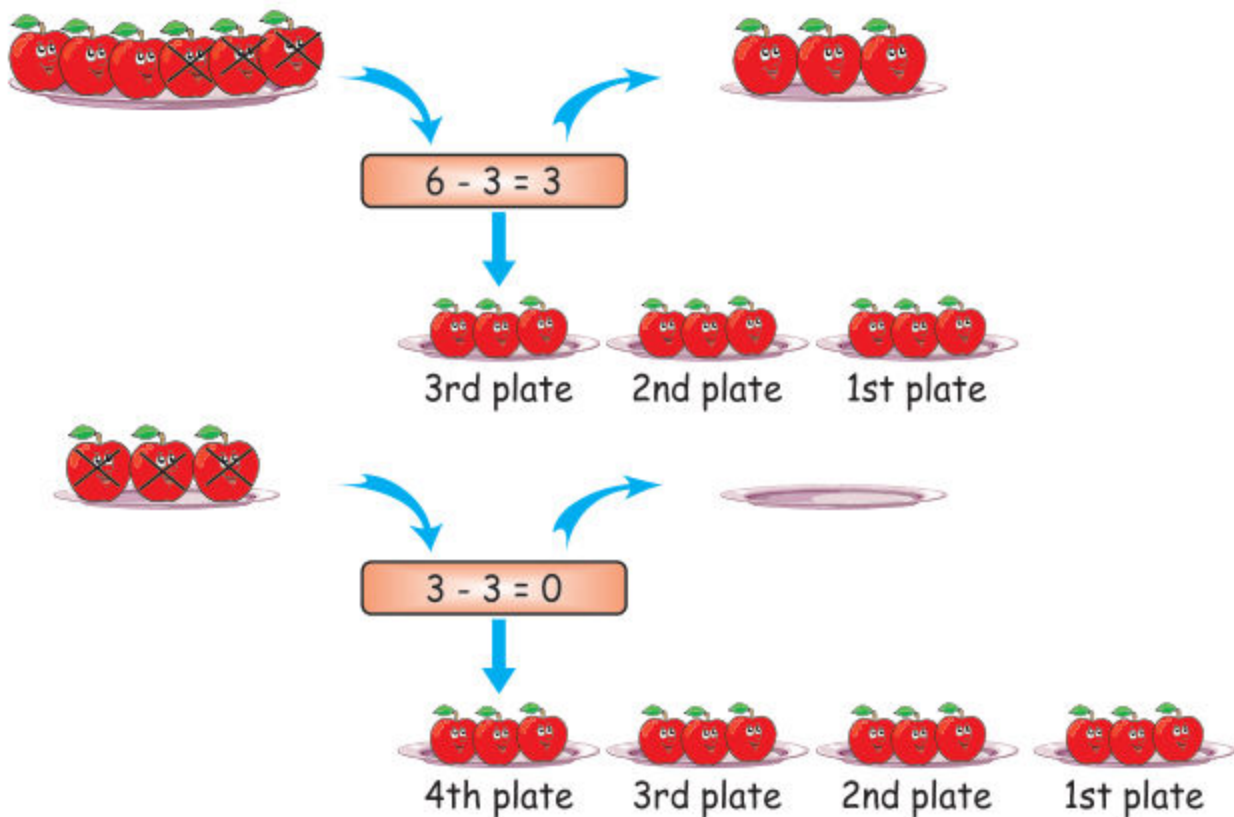
We know that multiplication is repeated addition. In the same way division is repeated subtraction as explained in the following example.

Example How many plates for 12 apples, do you need by putting 3 apples in each plate.

Solution: Let us subtract 3 apples repeatedly



Unit 2: Number Operations



We have got 4 plates of apples

In this process of division, we have subtracted 3 four times

So 12 divided by 3 is 4.

i.e. $12 \div 3 = 4$

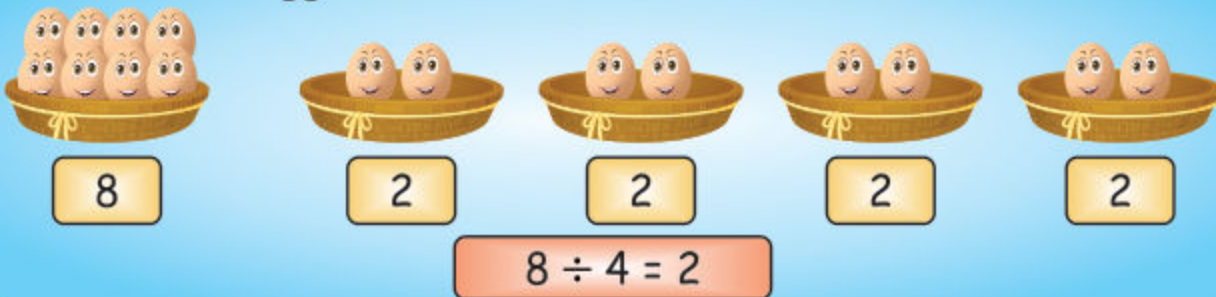
Note: We denote division by the symbol " $\div$ "

Recognizing that division of one number by another number can not be done in any order



Activity

Divide 8 eggs in 4 baskets.



If we try to divide 4 eggs in eight basket. Let us see what happens.



We can not divide 4 eggs in 8 baskets because eggs are less and baskets are more.

So $8 \div 4$ is not same as $4 \div 8$

Hence division can not be done in any order.

Exercise 11

- How many baskets for 15 toffees, do you need by putting 3 toffees in each baskets?



$$15 - 3 = 12$$



$$12 - 3 = \boxed{}$$



$$9 - \boxed{} = 6$$



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$$6 - 3 = \boxed{}$$



$$3 - \boxed{} = 0$$



$$\boxed{} \div \boxed{} = \boxed{}$$

So we need $\boxed{}$ baskets.

2. Write "T" for true and "F" for false

$6 \div 2$ is same as $2 \div 6$

$8 \div 2$ is not same as $2 \div 8$

$10 \div 2 = 2 \div 10$

$18 \div 6$ is not same as $6 \div 18$

$15 \div 3 = 3 \div 15$

Solving simple real - life word problem involving multiplication and division.

Example 1: Each student has 5 balls. How many balls do 4 students have.

Solution:



$$4 \times 5 = 20$$

So total number of balls = 20

Example 2:

How many packets for 30 toffees, do we need by putting 10 toffees in each packet?

Solution:



$$30 - 10 = 20$$



$$20 - 10 = 10$$



$$10 - 10 = 0$$



So $30 \div 10 = 3$

So we need 3 packets.

Exercise 12

1. A car has four wheels. How many wheels do 5 cars have?

Solution:



$$\boxed{4} + \boxed{4} + \boxed{4} + \boxed{4} + \boxed{4} = \boxed{5} \text{ times } \boxed{4}$$

$$= \boxed{20} \text{ wheels}$$

2. A bicycle has 2 wheels. How many wheels do 6 bicycles have?

Solution:

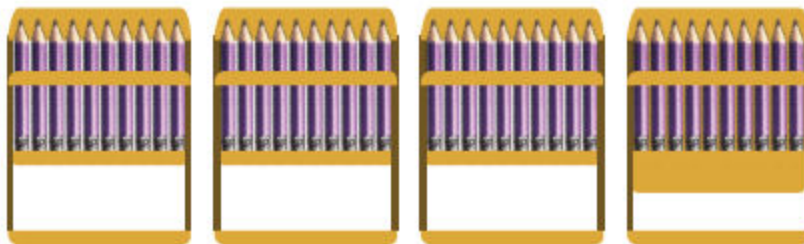


$$\boxed{} + \boxed{} + \boxed{} + \boxed{} + \boxed{} + \boxed{} = \boxed{} \text{ times } \boxed{}$$

$$= \boxed{} \text{ wheels}$$

3. A packet has 10 pencils. How many pencils are in 4 packets?

Solution:



$$\boxed{} + \boxed{} + \boxed{} + \boxed{} = \boxed{} \text{ times } \boxed{}$$

$$= \boxed{} \text{ pencils}$$

4. Ten books are divided equally among students. How many students do we need if each student gets 5 books?

Solution:



$$\square - \square = \square$$



$$\square - \square = \square$$



$$\square \div \square = \square, \text{so number of student} = \square$$

5. Twelve mangoes are divided equally in the plates. How many plates are needed if each plate has 4 mangoes.

Solution:



$$\square - \square = \square$$



$$\square - \square = \square$$



$$\square - \square = \square$$



$$\square \div \square = \square, \text{so number of plates} = \square$$

Unit

3

MONEY

Student Learning Outcomes:

At the end of the unit, Students will be able to:

- ✓ Identify Pakistani coins (Rs.1,2,5 and 10) and notes (Rs. 10, 20, 50, 100, and 500).
- ✓ Solve money problems involving addition and subtraction of Pakistani money.

